Hybrid Temporal Forecaster: A Combined Statistical and Deep Learning Approach for Energy Demand Forecasting

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***Abstract*—This report presents a hybrid temporal forecasting system for electricity demand prediction. Phase-1 focuses on exploratory data analysis and statistical modeling using ARIMA, while Phase-2 introduces a deep learning architecture to overcome its limitations. The proposed CNN-BiLSTM-Attention model significantly improves forecasting accuracy by capturing nonlinear dependencies and multi-scale seasonality.**

I. Introduction

Accurate short-term forecasting is critical in domains such as energy systems. Electricity demand exhibits complex characteristics including seasonality, noise, and regime shifts.

Phase-1 focused on statistical modeling, while Phase-2 extends this work using deep learning to improve predictive performance.

II. Literature Review

Classical time-series forecasting is based on the Box-Jenkins ARIMA framework [1]. Hyndman and Khandakar [2] introduced automated ARIMA selection. De Livera et al. [3] proposed TBATS for complex seasonality. The M4 competition [4] demonstrated that hybrid models outperform standalone approaches.

Recent advances in deep learning, particularly LSTM networks [5] and attention mechanisms [6], have significantly improved time-series forecasting performance. Architectures such as LSTNet [7] and Temporal Fusion Transformers [8] further enhance modeling of long- and short-term dependencies.

III. Dataset Description

The PJME hourly electricity dataset is used.

- Frequency: Hourly
- Duration: 2002–2018
- Target: PJME_MW

IV. Exploratory Data Analysis

The dataset shows strong seasonal behavior with daily and weekly cycles.

Rolling statistics indicate non-stationarity, suggesting the presence of underlying seasonal patterns.

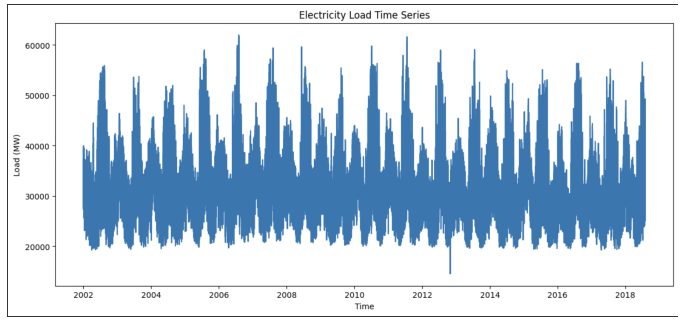

Fig. 1. Electricity Load Time Series

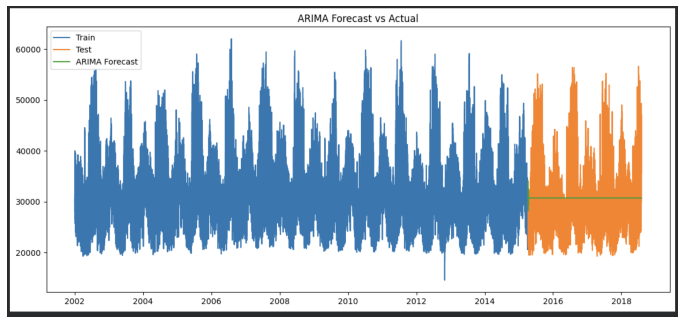

Fig. 2. ARIMA Forecast vs Actual

V. Stationarity Analysis

The Augmented Dickey-Fuller (ADF) test suggests stationarity, while the KPSS test indicates non-stationarity. This contradiction confirms that the dataset is **seasonally non-stationary**, requiring appropriate transformations for modeling.

VI. Baseline Model: ARIMA

The ARIMA model was implemented as a baseline statistical approach.

- MAE: 4986
- RMSE: 6498

VII. Limitations of ARIMA

ARIMA assumes linear relationships and struggles with complex seasonality and nonlinear dependencies. As a result, the model exhibits underfitting and fails to capture peak demand variations.

Model	MAE	RMSE
ARIMA	4986	6498
DL Model	1488	2045

TABLE I
PERFORMANCE COMPARISON

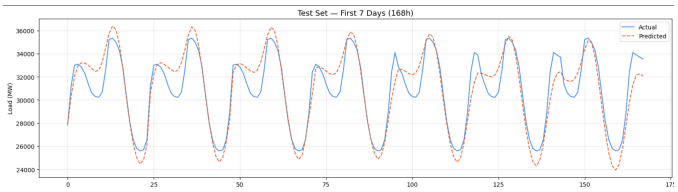

Fig. 3. Deep Learning Forecast vs Actual

VIII. PHASE-2: DEEP LEARNING MODEL

A. Data Pipeline

- Input window: 168 hours
- Output horizon: 24 hours
- Sliding window transformation
- RobustScaler normalization

This transformation enables the model to learn temporal dependencies across multiple time scales.

B. Model Architecture

The proposed architecture combines:

- Temporal CNN (kernel sizes 3 and 5) for local pattern extraction
- Bidirectional LSTM for long-term temporal dependencies
- Attention mechanism for dynamic feature weighting
- Fully connected layer for prediction

C. Attention Mechanism

$$e_t = v^T \tanh(W_1 H_t + W_2 h_{last}) \tag{1}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_k \exp(e_k)} \tag{2}$$

$$c = \sum_t \alpha_t H_t \tag{3}$$

D. Training Strategy

- Optimizer: Adam
- Loss: Huber Loss
- Learning rate scheduling
- Early stopping
- Dropout for regularization

IX. RESULTS

X. DISCUSSION

The deep learning model improves performance by:

- Capturing nonlinear relationships
- Learning long-term dependencies
- Dynamically focusing on important timesteps via attention

XI. CONCLUSION

The hybrid pipeline demonstrates that deep learning models outperform classical statistical models for complex time-series forecasting. The proposed architecture effectively captures multi-scale temporal patterns.

XII. REFERENCES

REFERENCES

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